

# THE STORY OF METALWORKING

FROM EVERYDAY OBJECTS TO COMPLEX MACHINES, METALS ARE VITAL FOR OUR CIVILIZATION

Since their earliest known use in the 8th millennium BCE, metals have played a crucial role in the production of a vast range of objects, and even today, with the availability of sophisticated polymers and composites, they still permeate every aspect of modern civilization.

Around 7000 BCE, naturally occurring metals, notably copper, began to be used for small items such as pins in western Iran and eastern Anatolia. These were made by simply grinding or beating the metal into shape. Heating copper to make it more malleable was probably discovered by accidentally dropping the metal in fire, but it was the introduction of smelting in a crucible around 3800 BCE that led to the large-scale use of metals.

## THE DEVELOPMENT OF ALLOYS

About 3000 BCE, the first alloy—bronze—was produced. Made by smelting tin and copper together in a crucible, bronze is stronger and more easily worked than either of its individual constituents, and it remained the principal metal for tools and weapons until the invention of ironworking around 1250 BCE. The technology to melt pure iron was not invented until the 19th century, so early iron objects were made by first smelting iron ore to an impure iron “bloom,” then separating out the iron pieces and welding them

together in a furnace. This method of production continued until the introduction of blast furnaces in Europe in the 15th century. The Industrial Revolution in the 18th century brought new techniques and the use of coking coal in blast furnaces, but it was English inventor Henry Bessemer’s invention of the Bessemer converter in 1856 that permitted the large-scale production of steel, a strong, high-quality, iron–carbon alloy. Later in the Industrial Revolution, further advances made it possible to produce other metals, such as aluminum, magnesium, and titanium, whose light weight and strength played a vital role in the development of the aviation and space industries.

**1,981°F**  
**THE MELTING POINT OF COPPER. WHEN COPPER IS ALLOYED WITH TIN, THIS DROPS TO 1,742°F**



## HOW ALLOYS ARE MADE

An alloy is a mixture of metals or of a metal with a nonmetal (such as iron with carbon in steel). Many metals occur naturally in alloyed form, but synthetic alloys were not produced until around 3000 BCE, when copper was melted with tin to produce bronze. The technique spread, reaching Mesopotamia soon after 3000 BCE and Egypt by 2000 BCE or possibly earlier.

### 700–800

#### Sword-making

In Europe, sword-makers develop stronger swords by welding together successive layers of iron with carbon added, or by beating out thin iron strips then welding them together.



Viking sword

### c. 15th century

#### Weapons from cast metal

Cast iron is developed. Because it is strong and can be used to make shapes such as tubes, it finds an immediate use in making artillery.



Medieval cannon



Lockheed Blackbird

### 1950s

#### Titanium aircraft

Because of its high strength-to-weight ratio, titanium starts to be used extensively in military aircraft. It is now also widely used in commercial aircraft.

### 800–1300/1450

#### Christian objects in precious metals

Medieval Christians make sacred objects, such as crucifixes and reliquaries, from gold and other precious metals, sometimes encrusted with gemstones.



The Verdun Altar

### 1810

#### Tin can

English inventor Peter Durand patents the tin can for preserving food. His patent was for a can made of iron and coated with tin to inhibit rusting of the iron.

### 1856

#### Bessemer converter

Englishman Henry Bessemer invents a converter that enables large-scale production of high-quality steel.



Bessemer converter

### 1910

#### Aluminum foil

The first aluminum foil is produced. It was made possible by the invention in 1886 of a method of mass-producing the metal by passing an electric current through molten ore.